

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ SOLUTIONS

Chapter 11: Geometric Progression

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 The general term

1. 1701 is the n th term of the G.P. 7, 21, 63, ... Find (a) n (b) the sum of these n terms.

ICSE 2025, 3 marks

SOLUTION

$$(a) a = 7, r = 3. 7 \times 3^{n-1} = 1701, \text{ so } 3^{n-1} = 243 = 3^5 \text{ and } n = 6.$$

$$(b) S_6 = \frac{a(r^6 - 1)}{r - 1} = \frac{7(729 - 1)}{2} = 7 \times 364$$

Answer: (a) $n = 6$ (b) sum = 2548

2. The 4th, 6th and last terms of a G.P. are 10, 40 and 640. The common ratio is positive. Find the first term, the common ratio and the number of terms. ICSE 2020, 4 marks

SOLUTION

$$ar^3 = 10 \text{ and } ar^5 = 40. \text{ Dividing: } r^2 = 4, \text{ so } r = 2.$$

$$a = \frac{10}{8} = \frac{5}{4}$$

$$\text{Last term: } \frac{5}{4} \times 2^{n-1} = 640, \text{ so } 2^{n-1} = 512 = 2^9 \text{ and } n = 10.$$

Answer: $a = \frac{5}{4}, r = 2, n = 10$

2 Sum of n terms

3. 15, 30, 60, 120, ... is a G.P. (a) Write its n th term. (b) How many terms add up to 945?

ICSE 2024, 4 marks

SOLUTION

$$(a) a = 15 \text{ and } r = 2, \text{ so } T_n = 15 \times 2^{n-1}.$$

$$(b) S_n = \frac{a(r^n - 1)}{r - 1} = 15(2^n - 1) = 945$$

$$2^n - 1 = 63, \text{ so } 2^n = 64 = 2^6.$$

Answer: (a) $T_n = 15 \times 2^{n-1}$ (b) 6 terms

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